

ANAS KHAN

FULL STACK ML ENGINEER

PYTHON · AGENTIC AI · LLMs · REACT.JS

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PORTFOLIO
anas-khan-portfolio-ak.vercel.app

SUMMARY

Full Stack ML Engineer with 5+ years of experience at Tassaract, where I moved up from Full Stack Developer to Full Stack ML Engineer and now lead a Python and AI team. I build complete web applications with Python (Django, Flask, FastAPI) and React.js, containerized with Docker and deployed on AWS. I design and ship machine learning and agentic AI systems, including multi-agent workflows built with LangGraph, and integrate LLM features into business applications. Comfortable owning a project end to end, from architecture to code to leading the people who build it.

SOFTWARE DEVELOPMENT EXPERIENCE

Team Lead – Python & AI Team

2024 – Present

TASSARACT

- Leading a team of Python and AI developers: project planning, task breakdowns, code reviews and on-time delivery.
- Designing and building full stack applications with Flask, Django, FastAPI, Node.js and React.js, containerized with Docker and deployed on AWS (EC2, S3, RDS).
- Building agentic AI systems and multi-agent workflows with LangGraph, and integrating LLM APIs (OpenAI, Groq) into products, including chatbot features.
- Using NumPy, Pandas and Matplotlib for data analysis and working with the team on machine learning models used in real business applications.
- Overseeing automation, web scraping and arbitrage bot projects, and mentoring junior developers.

Full Stack ML Engineer

2021 – 2024

TASSARACT

- Built full stack web applications with Python (Flask, Django), FastAPI and Node.js APIs and React.js, packaged with Docker.
- Wrote web scraping and automation scripts and bots, including early arbitrage bots, cutting down manual data entry.
- Built a chatbot using OpenAI's API to handle common customer and internal queries automatically.
- Delivered machine learning projects from data analysis to model deployment, and worked on blockchain wallet integration and on-chain transactions.

Full Stack Developer

2019 – 2021

TASSARACT

- Built web applications with Python (Flask, Django) and front-end pages with React.js, HTML, CSS and JavaScript.
- Wrote my first automation bots and scraping scripts, and assisted with early blockchain experiments; earned two promotions.

PROJECTS

ResearchCrew — Multi-Agent AI Research Assistant

Python · LangGraph · Groq · Streamlit · Plotly · ReportLab · SQLite

- Built a six-agent LangGraph pipeline that takes a topic, searches the web, reads and ranks sources, writes a cited report, verifies every claim against its source, and exports the result as a PDF with charts.
- **Agents:** Planner (focused search queries), Searcher (collects and ranks sources by credibility and relevance), Analyzer (extracts findings, decides if another search round is needed), Writer (citation on every factual sentence), Fact-Checker (checks each claim against its source and revises the report), Data Extractor (keeps only numbers found in the source text).
- Each report gets an Evidence Score from 0 to 100 plus a list of claims to double-check.
- Built the full app around the agents: user accounts, profile photo, per-user history in SQLite, light/dark mode and an admin panel. Built entirely with free tools.

Web3 Trading Agent

Python · LLMs · Multi-Agent AI · Tool Calling · Speech AI · Ethereum · Polygon · Solana

- Built a multi-agent AI assistant that turns natural language and voice commands into structured blockchain actions, with specialized agents for buying, selling, transfers and copy trading.
- Integrated three chains (Ethereum, Polygon, Solana), a live wallet view with balances, and a market watchlist.
- Built intent parsing and validation: a request like “Buy 5 USDC, then transfer 0.1” is split into two separate intents, and the agent asks for missing details (such as the target chain) instead of assuming.

Career Metro Mapping AI

Python · GPT-4o · Flask · MySQL · JWT · LLMs & Agents

- Built an app where a user uploads a CV and sees possible career paths drawn like a subway map, with a matching training plan exported as a PDF.
- GPT-4o generates the roadmap against a strict JSON schema so the front end can draw it every time; line colours, stations and branching rules are defined in the prompt, so any CV draws cleanly.
- Validation, retries and MySQL storage run in a background worker pool; the backend is built with Flask and JWT authentication.
- From a single role, the map extends to senior paths such as VP of Technology and Chief AI Officer.

AI Chatbot for Customer & Internal Queries

Python · OpenAI API · Flask / FastAPI

- Chatbot that automatically answers common customer and internal queries using OpenAI's API, reducing repetitive support work.

Arbitrage & Automation Bots

Python · APIs · scheduling tools · background jobs

- Arbitrage bots that track price differences across multiple platforms and act on profitable gaps automatically.
- Bots for data entry, monitoring, scraping and scheduled workflows, running reliably with minimal supervision.

Machine Learning & Data Analysis Projects

Python · NumPy · Pandas · Matplotlib · PyTorch

- Took ML projects from data analysis to building, deploying and integrating models into production web applications.

EDUCATION & CERTIFICATION

EDUCATION

Intermediate

Karachi Board

CERTIFICATION

AIC – Artificial Intelligence

Presidential Initiative for AI and Computing (PIAIC)

SKILLS

Languages	Python · JavaScript (ES6+) · HTML5 · CSS3
Agentic AI & LLMs	LangGraph · LangChain · Multi-Agent Systems · OpenAI API (GPT-4o) · Groq · Prompt Engineering · Structured Outputs (JSON Schema) · Tool Calling · RAG · Chatbots · Speech AI
ML & Data	NumPy · Pandas · Scikit-learn · SciPy · PyTorch · Matplotlib · Seaborn · Plotly
Backend	Django · Flask · FastAPI · Node.js · Express.js · REST APIs · JWT Auth · Celery · SQLAlchemy · Pydantic
Frontend	React.js · Streamlit · Responsive Design
Automation & Web3	Web Scraping (BeautifulSoup, Selenium) · Bot Development · Ethereum · Polygon · Solana
Databases	PostgreSQL · MySQL · SQLite · Pinecone (Vector DB) · ChromaDB (Vector DB)